ANATOMY OF AI AGENTS 2026

www.arianprabowo.com · 7-Layer AI Agent Stack Reference (Based on Roitman arXiv:2606.24937)

What actually runs under the hood of an autonomous AI agent in 2026?
Modern AI agents form a 7-layer left-to-right stack — from GPU silicon (L1) & uv/PyTorch/JAX (L2) to Data/RAG (L3), Arch/RLVR (L4), MCP (L5), Harness^2 (L6) & A2A swarms (L7).

① TRACE LEFT → RIGHT: Follow Layers 1 → 7 from silicon to swarms
② BUILD EACH LAYER: Top 3 pills = flagship 2026 primitives & tools
③ HARDEN PROD: Audit the red FAILURE MODES & TRAPS row at bottom

*Synthesized from Haggai Roitman's 'The Hitchhiker's Guide to Agentic AI: From Foundations to Systems' (arXiv:2606.24937, v1.4, June 2026) + 2026 systems tooling. Companion to AI Skill Tree 2026 & AI Ladder Evolution by Arian Prabowo · Updated: 2026-11-07 23:40:18 AEDT (UTC+11:00)

AI AGENT STACK

LAYER 1 → LAYER 7

Left-to-Right Pipeline
× 4 Architectural Rows

LAYER 1 →

PHYSICAL & SILICON

GPUs/TPUs/NPUs, HBM3e vs. SRAM hierarchy, CUDA/Triton kernels, and NVLink/InfiniBand fabrics.

- Core Unit: FLOPs, HBM Bytes & CUDA Warps
- Guide: Ch. 1.6, 2.1, 11.10–14, 30.5 (GPUs & Kernels).

LAYER 2 →

ENV, PACKAGES, PYTHON & TENSOR

Package managers (`Homebrew`, `uv`, `conda`, `pip`, `venv`), `Python`, `PyTorch`, `JAX`, `vLLM`, and sandboxes.

- Core Unit: Wheels, Lockfiles, Tensors & Paged KV
- Guide: Ch. 1.14–15, 11.1–9, 21.5, 22.1–9 (Env & Runtime).

LAYER 3 →

DATA, TOKENIZATION & RAG

Pre-training/SFT/RLVR data curation, synthetic trajectories, BPE tokenizers, vector DBs, and Agentic RAG.

- Core Unit: Token IDs, Embeddings & Chunks
- Guide: Ch. 1.2, 3.2–5, 4.3, 16, 18.4, 30.2 (Data & RAG).

LAYER 4 →

MODEL ARCHITECTURE & RLVR

Transformer/MoE/hybrid architectures, SFT, PEFT adapters (`LoRA`), preference tuning (`DPO`), and `RLVR` (`GRPO`).

- Core Unit: Logits, LoRA Adapters & CoT Traces
- Guide: Ch. 1, 2.2, 3–10, 12–13, 30.1–3 (Models & RL).

LAYER 5 →

CONTEXT, MEMORY, TOOLS & MCP

Context budgeting, 4-tier agentic memory (`Sleep-Time Compute`), function calling, `MCP`, and `SKILL.md`.

- Core Unit: Context Windows & JSON-RPC Calls
- Guide: Ch. 17, 18.2–3, 23, 24 (Memory, Tools, MCP & Skills).

LAYER 6 →

AGENT HARNESS, LOOP & GRAPH

The CPU/OS of an agent: `CLAUDE.md`/`AGENTS.md`, Loop Engineering, Graph Engineering (`LangGraph`), and `Harness²`.

- Core Unit: State Transitions, Actions & Diffs
- Guide: Ch. 18, 19, 20, 21, 27 (Harness, Loops & Graphs).

LAYER 7 ★

MULTI-AGENT MESH, A2A & UI

Multi-agent topologies (`OWC`), inter-agent protocols (`A2A`/`ACP`), parallel worktree swarms, and Agentic/Generative UI.

- Core Unit: A2A Tasks, Worktrees & GenUI Streams
- Guide: Ch. 21.6, 25, 26, 28, 30.10/18 (A2A, MAS & Agentic UI).

PILLAR 1

CORE PRIMITIVES & ARCHITECTURE

Hardware Physics,
Mathematical Ops,
Data Structures &
Control Topologies

- Layer 1–2: Silicon/Kernels
- Layer 3–4: Data/Model Math
- Layer 5–7: Memory/Loops/MAS

Layer 1

GPU / TPU / NPU Tensor Cores

HBM3e vs. On-Chip SRAM Hierarchy

FlashAttention IO-Aware SRAM Tiling

- Streaming Multiprocessors (SMs)
- Arithmetic Intensity (FLOP/Byte)
- FP16 / BF16 / FP8 / FP4 Formats
- Unified Memory Architecture (UMA)
- Warp Schedulers (32-Thread SIMT)
- Roofline Compute vs. Memory Bound
- Online Softmax Rescaling (Safe Max)
- Wafer-Scale & SRAM-Only LPU Dies

- Register File (256KB/SM)
- HBM3e Bandwidth (3.3–8 TB/s)
- Stochastic Rounding
- Memory Coalescing
- L1 Cache / Shared Memory
- SRAM Bandwidth (~20–30 TB/s)
- Tensor Memory Accelerator (TMA)
- Bank Conflict Avoidance
- L2 Cache (50MB–128MB)
- Transformer Engine FP8/FP4
- Systolic Array Matrix Multiply
- TDP Power & Liquid Cooling

Layer 2

Virtual Environments, Locks & Sandboxes

Autograd Computation Graphs & JIT

PagedAttention KV & Continuous Batch

- POSIX PTYs, Processes & Asyncio Loop
- OS Namespaces, cgroups & Seccomp
- PagedAttention Virtual Memory KV
- Speculative Decoding (Draft + Verify)
- ABI / manylinux Wheels & uv.lock
- Dynamic Autograd DAG & torch.compile
- RadixAttention Automatic Prefix Cache
- 3D Parallelism (DP/TP/PP/SP) & ZeRO

- pyproject.toml (PEP 517 / 621)
- LD_LIBRARY_PATH / DYLD RPATH
- XLA HLO Graph Fusion & CUDA Graphs
- KV-Cache FP8 / INT4 Quantization
- Content-Addressable Global Cache
- torch.compile (TorchInductor/Dynamo)
- Disaggregated Prefill-Decode Routing
- Container UnionFS / OverlayFS Layers
- Hardlink / Reflink Env Cloning
- JAX jit / vmap / pmap / shard_map
- Chunked Prefill & Guided Decoding
- Hardware Virtualization (KVM / HVF)

Layer 3

BPE Tokenization, Packing & Loss Masking

SFT, Preference & RLVR Data Curation

Embeddings, Vector HNSW & Hybrid RAG

- Byte-Level BPE / SentencePiece / Tiktoken
- Completion-Only Loss Masking (Labels=-100)
- Synthetic Reasoning & Trajectory Gen
- Hybrid RAG (Dense + BM25 + RRF Fusion)
- Sequence Packing & Position-ID Reset
- Chosen vs. Rejected Preference Pairs
- Bi-Encoder (HNSW) vs. Cross-Encoder
- GraphRAG Entity Summaries & REFRAG

- MinHash LSH & Exact Substring Dedup
- ChatML Special Tokens (<|im_start|>)
- Contextual Chunk Headers (Anthropic)
- Self-RAG Critique Tokens (ISREL/ISSUP)
- FastText / KenLM Quality Filtering
- Rejection Sampling (Best-of-N Filter)
- Late Interaction ColBERT Token MaxSim
- Corrective RAG (CRAG) Web Fallback
- Decontamination (N-gram / Embed Match)
- Execution-Verified Code/Math Traces
- Search-R1 RL Query Reformulation
- Matryoshka Representation (MRL)

Layer 4

Transformer · Sparse MoE · GQA / MLA

Hybrid Linear Attention (Mamba/DeltaNet)

PEFT (LoRA/DoRA) & RLVR (PPO/DPO/GRPO)

- Pre-Norm RMSNorm · SwiGLU · RoPE/YaRN
- Sparse MoE Top-K & Shared Experts
- LoRA · QLoRA (NF4) · DoRA · PiSSA
- Critic-Free RLVR: GRPO · Dr.GRPO · DAPO
- Grouped-Query (GQA) & Latent (MLA)
- Gated DeltaNet / Mamba-2 SSM Hybrid
- Preference Tuning: PPO · DPO · KTO
- Harness Distillation (Harness-Zero)

- Multi-Head Latent Attention (MLA)
- Zeroth-Order MeZO Forward-Only Tune
- RLCR Calibrated Brier Score Reward
- Cold-Start Pure-RL Emergence (R1-Zero)
- Aux-Loss-Free MoE Load Balancing
- Bradley-Terry RM & Process RM (PRM)
- On-Policy Self-Distillation (OPSD/RLCSD)
- MergeKit (SLERP / TIES / DARE)
- rsLoRA (alpha/sqrt(r)) · QLoRA NF4
- IPO · ORPO · SimPO · Online DPO
- Test-Time Compute (Best-of-N / MCTS)
- AdamW vs. Muon Matrix Orthogonal

Layer 5

Context Engineering & Token Budgeting

4-Tier Memory + Sleep-Time Dreaming

Tool Calling, MCP Bus & Progressive Skills

- Working Memory (Active Context Window)
- Semantic Memory (Entity/Fact Triples)
- Sleep-Time Offline Consolidation (Dream)
- Progressive Disclosure Skill Loading
- Episodic Memory (Past Experience Logs)
- Procedural Memory (SKILL.md / Weights)
- JSON-RPC 2.0 Host-Client-Server Bus
- Observation Masking & /compact Hygiene

- System / History / RAG / Tool Scratchpad
- Proactive Context Pre-Fetching
- Strict JSON Schema Tool Definitions
- Rolling Summary & Turn Truncation
- Write / Read / Reflect / Consolidate Ops
- KV-Cache Prefix Pinning (90% Discount)
- Code-Act (Python Execution as Tool)
- Subagent Forked Context Isolation
- Ebbinghaus Temporal Memory Decay
- Parallel Tool-Call Batch Execution
- Dynamic Tool Search (Top-K Schemas)
- Scratchpad & Todo.md State Offload

Layer 6

Agent Harness (Prompt+Tools+State+Loop)

Loop Engineering (Observe→Decide→Act)

Graph Engineering (Typed StateGraphs)

- ReAct · Reflexion · Plan-and-Execute
- Typed State Schema & Reducer Functions
- Fan-Out / Fan-In Parallel Send() API
- Spec-Driven Development (Spec→Plan→Task)
- 5 Loop Primitives (State/Step/Check/Stop)
- Conditional Edges & Cyclic Graph Routing
- Recursive Agent Harnessing (Harness^2)
- Cognitive Escalation & Uncertainty Handoff

- pi(a_t | s_t) Inference-Time Policy Loop
- BDI (Belief-Desire-Intention) Hybrid
- Stagnation & Repeat-Action Detection
- Exponential Backoff & Schema Self-Repair
- LATS (Language Agent Tree Search)
- Checkpointer Time-Travel & Replay
- Subagent Context Isolation (Forked Call)
- Token/Cost/Iteration Termination Caps
- LLM Compiler Parallel DAG Execution
- Human-in-the-Loop Interrupt() Gates
- Red-Green-Refactor TDD Agent Cycle
- Dynamic Model Routing (Fast vs. Reasoner)

Layer 7

Orchestrator-Worker-Critic (OWC) Pattern

Multi-Agent Topologies (Hierarchical/Mesh)

Generative UI & Shared State Sync

- Centralized Supervisor vs. Decentralized
- Debate / Society-of-Mind Consensus
- Contract-Net Task Bidding & Auctions
- A2A (Horizontal) × MCP (Vertical) Dual Bus
- Hierarchical Tree (Squads of Squads)
- Blackboard Shared-Memory Coordination
- Parallel Git-Worktree Swarm Isolation
- Human-on-the-Loop Escalation & Approval

- Agent Card (/.well-known/agent.json)
- Credit Assignment (Shapley / Counterfactual)
- Pub/Sub Asynchronous Event Choreography
- Optimistic UI Updates & Diff Previews
- Task Lifecycle (Submitted→Working→Done)
- Role Specialization & Persona Diversity
- Process-Centric Enterprise State Workflows
- Ambient OS Cross-App Delegation
- Multi-Agent MDP & CTDE MARL Training
- Byzantine Fault-Tolerant Critic Voting
- Server-Driven UI Component Streaming
- Organizational Digital-Twin Simulation

PILLAR 2

FRAMEWORKS, PKGS & PROTOCOLS

CLI Package Managers,
DL/Serving Frameworks,
Model Checkpoints &
Agent SDKs/Protocols

- brew · uv · conda · pip · venv
- PyTorch · JAX · vLLM · TRL
- MCP · SKILL.md · LangGraph · A2A

Layer 1

NVIDIA H100 / H200 / Blackwell B200

Google TPU v5p / v6e Trillium / v7

CUDA · OpenAI Triton · Apple Silicon M4

- NVIDIA GB200 NVL72 Rack Scale
- AWS Trainium2 & Inferentia2
- NVLink 4/5 (900–1,800 GB/s/GPU)
- cuBLAS / cuDNN / CUTLASS / FlashInfer
- AMD Instinct MI300X / MI325X
- Groq LPU & Cerebras WSE-3 ASIC
- InfiniBand NDR/XDR (400–800Gb/s)
- AMD ROCm / HIP & Apple Metal

- DGX / HGX 8-GPU Node Topology
- 80GB / 141GB / 192GB VRAM Tiers
- On-Device NPU (45–80+ TOPS)
- Liger-Kernel Fused Triton Ops
- GPUDirect RDMA / Storage
- Mac Mini M4 64GB-512GB Cluster
- Co-Packaged Silicon Photonics
- ECC Memory Error Correction
- Mellanox ConnectX-7 / 8 NICs
- RTX 4090 / 5090 Local Workstation
- Vera Rubin NVL144 (HBM4 Roadmap)
- NCCL / Gloo Collective Backend

Layer 2

Homebrew (brew) · uv · conda · pip · venv

Python 3.12+ · PyTorch 2.x · JAX · MLX

vLLM · SGLang · llama.cpp · Docker · E2B

- Homebrew (brew) & System Toolchains
- conda / Miniforge / mamba / Pixi / Poetry
- PyTorch 2.x · JAX / Flax · Apple MLX
- Docker / OCI & NVIDIA Container Toolkit
- uv (uv venv / uv run / uvx / uv pip)
- pip · venv · virtualenv · pipx
- vLLM · SGLang · TensorRT-LLM · Ollama
- Rust (PyO3/Maturin) & TypeScript/Bun

- brew install uv git ripgrep fd tmux
- pip install --no-build-isolation
- Outlines & XGrammar (JSON/CFG)
- Nix / Devcontainers / Direnv
- uv lock & uv sync --frozen
- DeepSpeed ZeRO & PyTorch FSDP2
- Ray Serve & NVIDIA Triton Server
- Slurm / Kubernetes (K8s) / Ray
- environment.yml & conda-lock
- llama.cpp (GGUF) & ExLlamaV2/v3
- ONNX Runtime & Safetensors Zero-Copy
- Pydantic v2 (Rust Schema Core)

Layer 3

HF Datasets · Datatrove · NeMo Curator

Tiktoken · SentencePiece · HF Tokenizers

Qdrant · Milvus · pgvector · Neo4j

- FineWeb-Edu · Dolma · The Stack v2
- Docling · Unstructured · Marker PDF
- BGE-M3 · Cohere Embed v3 & Rerank v3
- LlamaIndex · Haystack · LightRAG
- Apache Arrow · Parquet · WebDataset
- Tree-sitter AST Codebase Chunking
- FAISS · LanceDB · Chroma · Weaviate
- Distilabel · Curator Synthetic Suites

- Polars · DuckDB · PyArrow ETL
- OpenMathReasoning & NuminaMath
- Firecrawl · Jina Reader Markdown
- DVC & LakeFS Data Version Control
- Ray Data Distributed Streaming
- SWE-Gym & SWE-smith Trajectories
- Graphiti & Cognee Knowledge Graphs
- Cleanlab Automated Label Audit
- MosaicML StreamingDataset (MDS)
- ShareGPT & OpenAI Harmony Formats
- SQLite-vec & DuckDB-VSS Local
- Argilla Human-in-the-Loop Curation

Layer 4

GPT-5.4 / o3 · Claude 4.6/5 · Gemini 3

DeepSeek-R1/V3.2 · Qwen 3 · Llama 4

HF TRL · PEFT · verl · OpenRLHF · Unsloth

- TRL (SFTTrainer / DPOTrainer / GRPO)
- OpenRLHF (Ray + vLLM Distributed RL)
- Axolotl & LLaMA-Factory YAML Suites
- Prime Intellect verifiers & OpenR1
- verl (HybridEngine RLHF/RLVR at Scale)
- Unsloth (2x Speed Triton Backprop)
- HuggingFace Transformers & PEFT
- Torchtune & NeMo Aligner

- DeepSeek-R1-Distill (7B-70B Weights)
- OpenAI GPT-5.4-Codex & o4-mini
- bitsandbytes (NF4) · AWQ · GPTQ
- WandB / TensorBoard / MLflow Logs
- Qwen3-235B-A22B & QwQ-32B
- Gemini 2.5 / 3.1 Pro & 3.8 Flash
- Lean 4 / SymPy Formal Math Verifiers
- vLLM Rollout Worker Integration
- Claude Sonnet 4.5 / Opus 4.6 / 5.5
- Mistral Large 3 & Codestral
- MergeKit CLI Checkpoint Merger
- BitNet.cpp 1.58-Bit Ternary Weights

Layer 5

Model Context Protocol (MCP / .mcp.json)

Agent Skills (SKILL.md Folder Standard)

Letta (MemGPT) · Mem0 · Zep · LangMem

- MCP Primitives: Tools, Resources, Prompts
- FastMCP Python / TypeScript SDKs
- skills/scripts/ & skills/references/ Dirs
- Mem0 · Zep · Graphiti Temporal Memory
- MCP Transports: stdio & Streamable HTTP
- SKILL.md YAML Frontmatter + Instructions
- Letta (MemGPT) OS-Paged Memory Blocks
- Linux Foundation AAIF MCP Registry

- MCP Sampling & Roots Client Callbacks
- agentskills.io Open Specification
- Redis / Postgres Agent Session Stores
- GitHub / Slack / Postgres MCP Servers
- MCP Elicitation User Confirmation
- LangMem Semantic/Episodic Store
- OpenAPI-to-MCP Auto-Generators
- Anthropic & Cursor MCP Marketplaces
- .claude/skills/ & .agents/skills/ Paths
- Cognee & Memary Memory Engines
- Browserbase / Playwright MCP Servers
- Smithery & Glama MCP Registries

Layer 6

Claude Code CLI · Codex CLI · OpenCode

Cursor · Windsurf · Cline / Roo · Aider

LangGraph · OpenAI Agents SDK · DSPy

- CLAUDE.md & AGENTS.md Repo Constitutions
- GitHub Spec Kit & obra/superpowers TDD
- Smolagents (CodeAct Python-as-Action)
- OpenHands (OpenDevin) & Block goose
- .cursor/rules & .github/copilot-instructions
- LlamaIndex Workflows (Event-Driven)
- PydanticAI Type-Safe Agent Harness
- Semantic Kernel & Haystack Agents

- LangGraph PostgresCheckpointer
- Browser-Use & Playwright DOM Harnesses
- Instructor & BAML Structured Outputs
- Repomix / Gitingest Codebase Packers
- DSPy MIPROv2 Prompt/Weight Compiler
- Vercel AI SDK Core (generateText/maxSteps)
- Litellm Unified Model Router
- LangSmith · Arize Phoenix · Langfuse
- Claude Agent SDK & Computer Use API
- Guardrails AI & NeMo Guardrails
- Git Pre-Commit & CI Harness Hooks
- OpenTelemetry (OTel) GenAI Spans

Layer 7

Google A2A Protocol · IBM ACP · ANP

CrewAI · AutoGen / AG2 · LangGraph Swarm

CopilotKit · AG-UI · Vercel AI SDK UI

- Google Agent-to-Agent (A2A) JSON-RPC/SSE
- Microsoft AutoGen v0.4 / Semantic Kernel
- Claude Code Subagents & Worktree Swarms
- Agent Gateways (Kong / Solo / LiteLLM)
- IBM BeeAI & Agent Communication Protocol
- CrewAI Flows & Role-Based Crews
- OpenClaw 24/7 Multi-Channel Agent Daemon
- Claude Artifacts · ChatGPT/Gemini Canvas

- AG-UI (Agent-User Interaction Protocol)
- OpenAI ChatKit & Realtime WebRTC Voice
- OpenAI Swarm / Agents SDK Handoffs
- Slack / Discord / WhatsApp Bot Bridges
- Vercel AI SDK useChat / streamUI Hooks
- LangGraph Platform Cloud & Studio UI
- A2A Artifact & Push-Notification Webhooks
- Linear / GitHub PR Async Agent Teammates
- Chainlit · Streamlit · Gradio Agent UIs
- Camel-AI Workforce & MetaGPT SOPs
- React Server Components (RSC) GenUI
- 1-Click MCP & A2A Enterprise Registries

PILLAR 3

SCALING, EVALS & SECURITY

Throughput SLAs,
MicroVM Sandboxes,
Verification Oracles &
Zero-Trust Guardrails

- MFU · E2B/bwrap · RAG Triad
- RLVR Graders · MCP OAuth 2.1
- SWE-bench · pass^k · KYA

Layer 1

Model FLOPs Utilization (MFU > 45%)

Prefill (Compute) vs. Decode (BW)

Confidential Compute GPU TEEs

- Prefill Phase (Matrix-Matrix Gemm)
- Multi-Instance GPU (MIG) Slices
- NVIDIA DCGM Hardware Health Check
- Nsight Compute & Systems Profiling
- Decode Phase (Matrix-Vector Gemv)
- Rail-Optimized Fat-Tree Topology
- Silent Data Corruption (SDC) Scans
- Hardware Root-of-Trust Attestation

- MFU = Achieved / Peak FLOPs
- 4-Model RLHF VRAM Multiplier
- NVLink All-Reduce Bandwidth
- H100/B200 Secure Enclave (CC-On)
- Hardware FLOPs Utilization (HFU)
- Optimizer State Memory (12B/param)
- Cross-Node All-to-All Bottleneck
- Power Usage Effectiveness (PUE)
- Bytes/Param: 2B (BF16) vs 0.5B (FP4)
- Activation Memory Checkpointing
- Thermal Throttling Mitigation
- Straggler GPU Detection

Layer 2

E2B / Firecracker MicroVM Sandboxes

OS Sandboxing (Bubblewrap / Seatbelt)

TTFT / TPOT SLAs & Async RL Sync

- gVisor User-Space Kernel Isolation
- OpenEnv Standardized Gym Reset/Step
- Scoped Ephemeral OAuth / API Tokens
- Per-Tenant LoRA Adapter Hot-Swapping
- macOS sandbox-exec & Linux bwrap
- Air-Gapped VLAN & Egress Firewall
- NCCL Weight Broadcast (Trainer->vLLM)
- Grammar-Constrained 100% Valid JSON

- env.reset() / env.step(action) API
- TTFT < 200ms · TPOT < 15ms/tok
- S-LoRA / Punica Multi-Adapter Batch
- SBOM & pip-audit Vulnerability Scan
- SWE-bench Docker Repo Images
- Goodput (SLO-Compliant Tok/s)
- Pyodide / Wasm Code Execution
- Signed Container Attestation (Cosign)
- OSWorld / WebArena Live Desktop VMs
- In-Flight Weight Update (Zero Stall)
- Secret Scrubbing in Env Vars
- Deterministic Seed Mode (cuDNN)

Layer 3

RAG Triad Evals (Faithfulness / Recall)

Benchmark Decontamination & PII Scrub

Document-Level RBAC & Injection Scans

- Ragas & TruLens Groundedness Oracles
- Pre-Indexing ACL / RBAC Permission Tags
- Canary String & N-Gram Contamination
- Citation Span-Locking & Provenance
- NDCG@k · MRR · Recall@k Retrieval Audits
- Regex + Presidio PII & Secret Scrubbing
- Synthetic Data Diversity & Entropy Audit
- Indirect Prompt Injection Sanitization

- LLM-as-a-Judge Data Quality Scoring
- Needle-in-a-Haystack (NIAH) & RULER
- Poisoned Document Backdoor Scans
- Embedding Drift & Stale Index Check
- Perplexity Outlier Filtering (KenLM)
- FRAMES & LongBench Multi-Hop Eval
- Copyright & SPDX License Filtering
- Tokenizer Fertility (Tok/Word) Audit
- Topic & Difficulty Curriculum Mix
- Hallucination Span (LettuceDetect)
- Cross-Tenant Vector Namespace Auth
- Zero-Retention Enterprise Pipelines

Layer 4

Verifiable Execution Oracles (RLVR)

KL-Divergence & Entropy Regularization

Constitutional AI & Safety Classifiers

- Deterministic Unit-Test / Math Graders
- Multi-Objective Reward Combination
- Instruction Hierarchy Privilege Training
- Activation Steering & SAE Circuit Probes
- DAPO Clip-Higher & Dynamic Sampling
- Listwise Rank-Based Reward Models
- Deliberative Alignment in CoT Traces
- GCG / AutoDAN / Best-of-N Red-Teaming

- KL Penalty beta * D_KL(pi_theta || pi_ref)
- Pass@k vs. Majority@k Scaling
- Sleeper Agent Backdoor Detection
- CBRN / Cyber ASL-3/4 Threshold Gates
- PPO Clipping epsilon in [0.1, 0.28]
- Length-Controlled Win-Rate Audits
- Sandbagging & Alignment Faking Audits
- Contamination-Free LiveBench / HLE
- GRPO Group Size G = 8..64 Rollouts
- Catastrophic Forgetting Evals
- Machine Unlearning (RMU / Grad Ascent)
- Calibration Expected Error (ECE)

Layer 5

MCP OAuth 2.1 Scopes & Tool Signing

/compact Hygiene (<65% Context Cap)

Memory Contradiction & Decay Audits

- Human-in-the-Loop Destructive Tool Gate
- Memory Contradiction & Stale-Fact Purge
- MCP Server Code-Signing Verification
- OX Security stdio RCE Hardening
- Tool Schema Token-Bloat Minimization
- LOCOMO & LongMemEval Memory Evals
- Least-Privilege Read-Only Connectors
- Cross-Tenant Memory Isolation

- Context Utilization < 65% Rot Threshold
- ToolBench & API-Bank Multi-Tool Eval
- Skill Static Analysis Before Install
- Idempotency Keys for Mutating Tools
- Lost-in-the-Middle U-Curve Placement
- Episodic Replay Buffer Deduplication
- Egress Domain Whitelisting per Tool
- Audit Logging of All JSON-RPC Calls
- BFCL (Berkeley Function-Calling) Eval
- Tool Output Truncation & Pagination
- OAuth 2.1 PKCE Token Refresh Flow
- Canary Tokens in Memory Stores

Layer 6

Decoupled Verification Oracles (No Self-Grade)

SWE-bench Verified/Pro & Terminal-Bench

pass^k Reliability & Trajectory Tracing

- Property-Based Testing (Hypothesis/Kani)
- Immutable CI Tests Locked From Agent Write
- LLM-as-a-Judge Pairwise Rubric Grading
- PyRIT & Garak Adversarial Red-Teaming
- Mutation Testing (Stryker / mutmut) Gates
- Trajectory Tool-Call Exact/Prefix Match
- Cost-per-Resolved-Task Efficiency Metric
- LlamaFirewall Prompt/Tool Guardrails

- pass^k = (pass@1)^k Consistency Metric
- Scale SWE-bench Pro (Private Repos)
- Shadow-Mode Traffic Counterfactual Eval
- Git Diff AST Security Linter (Semgrep)
- METR 50%-Reliability Horizon (2h-30h)
- Step Efficiency (Optimal / Actual Steps)
- Circuit-Breaker Spend & Loop-Depth Caps
- Automated Rollback on Regression
- GAIA · OSWorld · WebArena · TAU-bench
- Deterministic Mock-Tool Replay Harness
- Canary Token Exfiltration Tripwires
- EU AI Act / SOC2 Immutable Audit Logs

Layer 7

Agent Gateway (mTLS, Auth, FinOps Caps)

Cross-Agent Trust & Cryptographic KYA

Independent Critic & Red-vs-Blue Swarms

- Zero-Trust Agent Identity (SPIFFE / KYA)
- Per-Swarm Token & Dollar Circuit Breakers
- Cross-Repo API Contract Locks (OpenAPI)
- Provenance Watermarking (C2PA / SynthID)
- OAuth 2.1 On-Behalf-Of (OBO) Token Flow
- Anti-Collusion & Decorrelated Critic Models
- Deadlock, Livelock & Cycle Detection
- Strict Liability & Action Audit Ledgers

- Communication Overhead O(N^2) Pruning
- Swarm Amplification Factor Monitoring
- XBOW / Cybench / T3MP3ST Cyber Ranges
- Anti-Sycophancy & Anti-Psychosis Filters
- Multi-Agent Judge Independence Check
- Canary Deployment of Agent Topologies
- Lean 4 / TLA+ Formal Protocol Verification
- GDPR / HIPAA Cross-Agent Consent Scopes
- Cross-Org Data-Loss Prevention (DLP)
- Chaos Engineering (Injecting Rogue Agents)
- Human Escalation SLA Latency Tracking
- Kill-Switch Broadcast Across Swarm Nodes

ANTI-PATTERNS

FAILURE MODES & TRAPS

What Breaks in Prod:
Bottlenecks, Collapses,
Injections & Swarm
Echo Chambers

- OOM & CUDA ABI Hell
- Model Collapse & Reward Hack
- Context Rot & O(N^2) Chatter

Layer 1

HBM Memory Bandwidth Wall (Decode)

CUDA Out-of-Memory (OOM) Crashes

Silent Tensor Core Bit-Flip Corrupt

- ▸ Low Decode MFU (<5% Compute Util)
- ▸ PCIe Host-to-Device Bottleneck
- ▸ InfiniBand Link Flaps & Stalls
- ▸ Straggler GPU Halting 1,000 Nodes

- 4-Model RLHF VRAM Explosion
- Unaligned Tensor Shapes (No TC)
- Thermal Throttling Under RL Rollouts
- Consumer GPU P2P Disabled Locks

Layer 2

Python Dependency & CUDA ABI Hell

Unsandboxed Host RCE & rm -rf /

KV-Cache Fragmentation & Preemption

- ▸ nvcc vs. torch.version.cuda Mismatch
- ▸ Global System Python Pollution
- ▸ pip install Breaking flash-attn ABI
- ▸ Typosquatted PyPI / npm / Skill Packs

- Agent Leaking .env & SSH Keys
- RL Rollout Generation Bottleneck
- 60-80% Contiguous KV Memory Waste
- torch.compile Graph-Break Thrashing

Layer 3

Garbage-In Synthetic Model Collapse

Naive Fixed-Size Mid-Table Chunking

Indirect Prompt Injection via RAG

- ▸ Training on Prompt Tokens (No -100 Mask)
- ▸ Test-Set Contamination Mirage
- ▸ Cross-Row Table & Function Splitting
- ▸ Unfiltered Vector RAG Leaking Docs

- Chat Template EOS/BOS Token Mismatch
- Unverified Synthetic Hallucinations
- Tokenizer Glitch & Byte-Fallback Bloat
- Hallucinated Citations on Unread Text

Layer 4

Reward Hacking & Length Inflation

Entropy Collapse & Mode Collapse

Catastrophic Forgetting / Alignment Tax

- ▸ GRPO Zero-Variance Advantage (All 0/1)
- ▸ DPO Likelihood Displacement (Both Drop)
- ▸ MoE Routing Collapse (Expert Starve)
- ▸ Naive INT4 Quant Breaking Reasoning

- Sycophancy & Delusional Validation
- Over-Refusal on Benign Triggers
- Emergent Misalignment from Narrow SFT
- FP16 Loss-Scaling Underflow / NaNs

Layer 5

Context Rot & Lost-in-the-Middle

MCP Tool Bloat (100+ Schemas)

Malicious Skills & MCP Rug-Pulls

- ▸ 100+ MCP Tool Schemas Blowing Context
- ▸ ClawHavoc Malicious SKILL.md Payloads
- ▸ MCP Rug-Pull & Tool Shadowing Attacks
- ▸ Memory Poisoning Across Sessions

- Stale Episodic Memory Overriding Code
- Malformed Tool-Call JSON Loops
- stdio MCP Server Spawning Host Shells
- Unbounded Tool Output Flooding Window

Layer 6

Agent Deleting / Mocking Failing Tests

Unbounded Loop Burn & Action Stagnation

God-Node Anti-Pattern & State Bloat

- ▸ Coupled Agent Rewriting Its Own Grader
- ▸ 0.95^20 = 35.8% Compounding Step Decay
- ▸ Poisoned .cursor/rules or GitHub Issues
- ▸ CVE-2025-53773 IDE Config Auto-Approve RCE

- Swallowing Exceptions in Tool Wrappers
- Premature Termination Before Verification
- Appending Raw 50KB HTML to Graph State
- Sycophantic Plan Pivot on User Pushback

Layer 7

Cascading Hallucination & Swarm Echo

O(N^2) Chatty Agent Token Explosion

Cross-Agent Confused Deputy & Escalation

- ▸ 7-Persona Roleplay Chatter Burning 15x Cost
- ▸ Homogeneous Critic Blindly Agreeing
- ▸ Supervisor Bottleneck & Context Overflow
- ▸ Merge Conflicts Across Parallel Worktrees

- Rogue Subagent Bypassing Parent Scopes
- Free-Riding Worker Agents in Crews
- Infinite Ping-Pong Handoff Deadlocks
- UI Stream Desync & Phantom Tool States